**Career Mobility and Scientific Impact: A Reproduction Study Using OpenAlex**


## Abstract

Scientific mobility is common in academia, yet its impact on research performance remains unclear. This study reproduces part of the analysis from *Career on the Move* [1] using OpenAlex data to examine how changes in institutional rank affect scientific impact. A sample of 651 physicists and ~55,600 publications was analyzed. Institutional rank differences (Δr) and performance changes (Δc*) were calculated for each move. Results show that moving to lower-ranked institutions decreases performance, while moving to higher-ranked institutions shows a small but consistent increase. These findings partially reproduce the original study but differ in observing a slight gain when moving to higher-ranked institutions. Overall, institutional environment has an asymmetric influence, with a stronger negative than positive effect.


## Introduction

Academic careers often involve moving between institutions, and these moves are commonly believed to influence research performance. Higher-ranked institutions typically provide more funding, resources, and collaboration opportunities, leading to the assumption that moving to elite institutions improves scientific impact. However, it is unclear whether these advantages translate directly into better performance. Previous research has shown that scientific mobility follows structured patterns and may not always lead to improved outcomes [1]. Understanding this relationship is important because institutional prestige plays a significant role in hiring, promotion, and evaluation decisions. This project examines whether scientific success is primarily driven by institutional environment or by individual researchers.

## Goal and Hypothesis

The goal of this study is to examine how changes in institutional ranking affect a researcher's scientific impact using a different dataset. The hypothesis is that if institutional prestige strongly influences productivity, then moving to higher-ranked institutions should increase scientific impact, while moving to lower-ranked institutions should decrease it [1]. The null hypothesis is that scientific performance is mainly determined by the individual, and therefore institutional moves will not significantly affect impact.

## Methods

Data were obtained from the OpenAlex database [2] and filtered to include physics-related publications between 2000 and 2020. A random sample of 1,000 authors was collected, and after filtering for sufficient publication history and institutional information, 651 authors were retained. The final dataset included approximately 55,600 publications and around 1,000 institutions.

Institutional ranking was determined using the total number of citations associated with each institution, consistent with prior work [1]. Total citation counts were used as a proxy for 5-year citations due to data limitations. For each author, career trajectories were reconstructed using publication affiliations, grouping consecutive publications at the same institution into "stays."

For each move between institutions, the change in rank was calculated as Δr = destination rank − origin rank. Positive Δr indicates movement to a lower-ranked institution, while negative Δr indicates movement to a higher-ranked institution. Scientific performance was measured using Δc*, defined as the standardized difference between average citation counts before and after the move [1]. Moves were grouped into bins based on Δr, and average Δc* values were calculated for each bin.

**Results**

Figure 1 shows the relationship between Δr and Δc*. The x-axis represents the change in institutional rank, while the y-axis represents the change in scientific performance. Negative Δr values correspond to moves to higher-ranked institutions, while positive values correspond to moves to lower-ranked institutions.

The results show a clear downward trend. When Δr is positive, indicating movement to lower-ranked institutions, Δc* becomes increasingly negative, demonstrating a decline in performance. In contrast, when Δr is negative, representing movement to higher-ranked institutions, Δc* shows a small but consistent positive increase.

Importantly, the relationship is not symmetric, as performance decreases more strongly when moving to lower-ranked institutions than it improves when moving to higher-ranked institutions. While the overall downward trend is consistent with previous findings [1], the presence of a small positive increase suggests a potentially stronger relationship between institutional ranking and performance in this dataset. This differs slightly from the original study, which reported no improvement when moving to higher-ranked institutions, whereas a small positive effect is observed here.

**Discussion**

The results partially reproduce the findings of the original study. Consistent with prior work, moving to lower-ranked institutions is associated with a decrease in scientific impact [1]. However, unlike the original study, which reported no gain in impact when moving to higher-ranked institutions, this analysis shows a small but consistent positive effect.

This suggests a modest positive influence, alongside a stronger negative effect. Overall, the results indicate an asymmetric relationship, where institutional environments constrain performance more strongly than they enhance it. While the positive effect remains relatively small, it may reflect the combined influence of improved resources, collaborations, and research conditions at higher-ranked institutions.

Several limitations should be considered. First, the dataset is smaller than the original study, which may introduce variability. Second, OpenAlex data differs from the APS dataset used in the original paper, potentially affecting career reconstruction and citation counts. Third, total citations were used instead of 5-year citations, which may introduce bias, as OpenAlex does not provide citation counts within a fixed time window. Finally, simplified filtering and binning methods may not fully capture the complexity of the original analysis.

Despite these limitations, the overall trend remains consistent, suggesting that institutional prestige alone does not determine scientific success.

**Conclusion**

This study reproduced the relationship between institutional mobility and scientific performance using OpenAlex data. Consistent with the original study, moving to lower-ranked institutions leads to a clear decline in performance. However, in contrast to the original findings, a slight increase in performance is observed when moving to higher-ranked institutions.

These results suggest that institutional environment has an asymmetric influence on scientific impact, with a stronger negative effect when moving to lower-ranked institutions than the positive effect observed when moving to higher-ranked institutions. Overall, both institutional context and individual ability contribute to scientific success, with institutions playing a modest but measurable role.

**Appendix**

**Figure 1**

*Relationship between institutional rank change (Δr) and performance change (Δc*) using OpenAlex data.** Each point represents a binned average of career moves, and error bars indicate standard error. The plot shows that moving to lower-ranked institutions (positive Δr) is associated with a decrease in performance, while moving to higher-ranked institutions (negative Δr) is associated with a slight increase in performance.




**References**

**[1]** Deville, P., Wang, D., Sinatra, R., Song, C., Blondel, V. D., & Barabási, A.-L. (2014). *Career on the move: Geography, stratification, and scientific impact*. Scientific Reports, 4, 4770.

**[2]** OpenAlex. (2023). *OpenAlex: An open catalog of scholarly papers, authors, and institutions*. https://openalex.org